Yuyan He

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I work in the subfield of mathematical logic called model theory, and I am particularly interested in its interaction with combinatorics.

Education

University of California, Los Angeles
B.S. in Mathematics *2018-2022*
Honors: Summa Cum Laude, departmental highest honor

University of Notre Dame
Ph.d. candidate in Mathematics *2022–present*
Advised by Sergei Starchenko

Research

Preprints

Sum-product Phenomenon Via Dimension: with Sergei Starchenko

The Composition Lemma for n-dependence: with Artem Chernikov

Select Talks

Candidacy exam **University of Notre Dame**
Pseudo-finite dimensions and linear algebraic groups *Feb 2024*
Hrushovski's treatment of approximate subgroups of linear groups

ASL NA Annual Meeting Model-Theoretic Algebra Special Session **University of Pennsylvania**
Sum-product phenomenon via dimension *July 2026*
On joint work with Sergei Starchenko

AMS Fall Sectional Meeting Model Theory Special Session **University of Minnesota, Duluth**
Sum-product phenomenon via dimension *Oct 2026*
On joint work with Sergei Starchenko

Teaching

As Teaching Assistant at University of Notre Dame
Fall 2023: Math 10350 Calculus A

Spring 2024: Math 10560 Calculus II

Fall 2024: Math 10450 Glynn Mathematics Seminar I

Spring 2025: Math 10860 Honors Mathematics II

Fall 2026: Math 12150 Principles of Calculus

As Lecturer at University of Notre Dame Fall 2026: Math 12150 Principles of Calculus

Fall 2025: Math 12560 Calculus II

Funded Workshops

UNIMOD 2022
University of Leeds *July-August 2022*
Schools and workshop in model theory

Fudan Logic Summer School
Fudan University *August 2023*

Graduate Workshop in Logic and Combinatorics
University of Illinois, Chicago *August 2026*

Service

Graduate Student Logic Seminar **Co-organizer**
University of Notre Dame *2023-2024*

Graduate Student Conference in Logic **Co-organizer**
University of Notre Dame *Spring 2025*

Awards

2022: Daus Memorial Award, Universilty of California, Los Angeles
2023,2024,2026: ASL Student Travel Award